

Fig. 26.7. Left: Matthew Choptuik. *Middle*: An imploding gravitational wave. *Right*: The boiling produced by the wave, and the naked singularity at the center of the magnifying glass.

Now, such a singularity can never occur in nature. The required tuning is not a natural thing. But an exceedingly advanced civilization could produce such a singularity artificially by precisely tuning a wave's implosion, and then could try to extract the laws of quantum gravity from the singularity's behavior.

Upon seeing Choptuik's simulation, Stephen conceded our bet—"on a technicality," he said (bottom of Figure 26.6). He thought precise tuning unfair. He wanted to know whether naked singularities can occur *naturally*, so we renewed our bet with a new wording that the singularity must arise without any need for precise tuning. Nevertheless, Stephen's concession, in a very public venue (Figure 26.8) was a big deal. It made the front page of the *New York Times*.